

Data Engineering Project

1. Title: End-to-End E-Commerce Data Platform for Sales and Customer Analytics

2. Group Members:

- **Vishwa Narayanaswamy:**

Responsible for **data collection and data cleaning**. Prepares raw data, handles preprocessing, and ensures data quality before it enters the pipeline.

- **Sheerin Banu Shiek Mohamed:**

Responsible for **pipeline architecture and ETL workflows**. Builds and manages Airflow pipelines, Docker setup, deployment, and system monitoring.

- **Koushik Raj Singh:**

Responsible for **data analysis, warehouse design, and dashboard creation**. Designs SQL queries, builds warehouse tables, performs analysis, and develops dashboards for insights. Also contributes to documentation and testing.

3. Problem Statement:

E-commerce businesses generate large amounts of data, but without a proper system it is hard to use effectively. This project builds an automated platform to organize data and support smarter decisions.

4. Introduction:

E-commerce has become a major part of everyday life, with people shopping online for convenience and variety. Every search, click, and purchase creates valuable data that can help businesses understand customer behavior, track sales, and improve services.

However, in many organizations, data is stored in different systems such as order databases, customer records, and product catalogs. Because the data is scattered, it becomes difficult to combine and analyze it quickly. Many businesses still rely on manual reports using spreadsheets, which takes time and can lead to errors. As the amount of data increases, this approach becomes slow and inefficient.

This project focuses on building an End-to-End E-Commerce Data Platform to solve these problems. The system automatically collects data from multiple sources, cleans and transforms it, and stores it in an organized way. The processed data is then presented through dashboards and

reports that show insights like sales trends, customer behavior, and product performance. By automating the entire process, the platform ensures accurate and up-to-date information.

Overall, this project shows how data engineering can turn raw data into useful insights that help businesses improve performance and make better decisions.

5. Project Execution

The project execution plan outlines the complete workflow and timeline for developing the End-to-End E-Commerce Data Platform for Sales and Customer Analytics. It helps organize project activities, manage task dependencies, track milestones, and ensure the project is completed efficiently and on schedule

- Gantt Chart

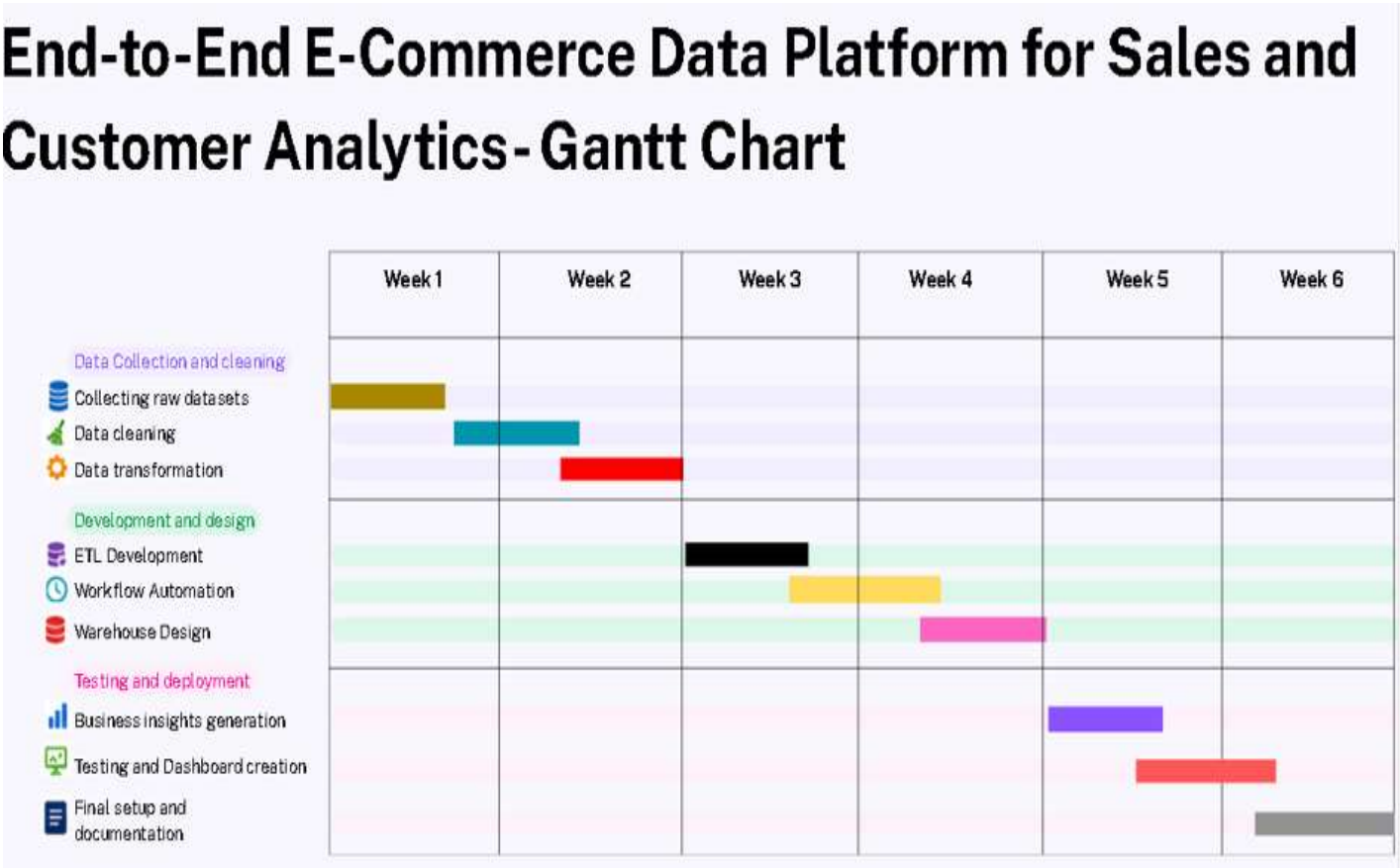


Fig 1: Gantt Chart

The Gantt chart gives a simple and clear view of how the End-to-End E-Commerce Data Platform for Sales and Customer Analytics project will be completed over 6 weeks. It shows when each major task,

such as data collection, cleaning, ETL pipeline development, analytics, dashboard creation, testing, and deployment, will be carried out. This helps the team stay organized, track progress easily, and make sure the project is completed smoothly and on time.

- **Activity Flow Diagram**

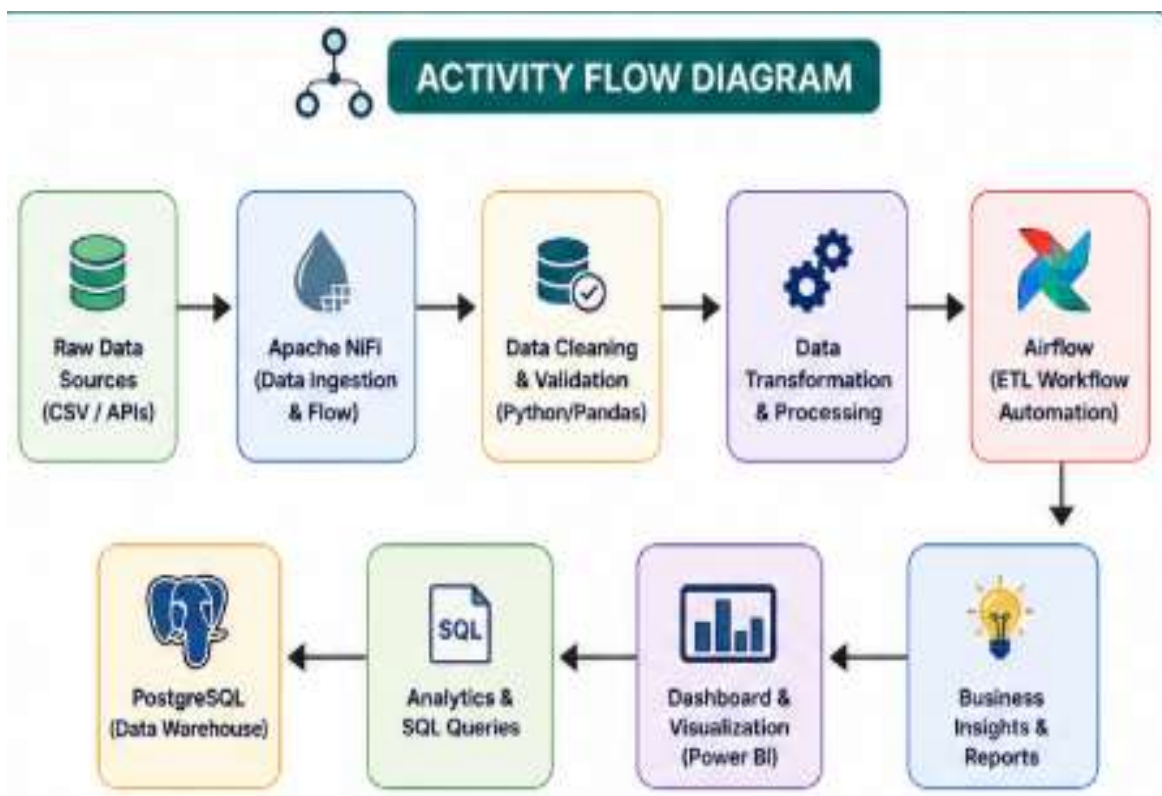


Fig 2: Activity Flow Diagram

The activity flow diagram illustrates the step-by-step workflow of the End-to-End E-Commerce Data Platform for Sales and Customer Analytics. It shows how raw e-commerce data is collected, processed, transformed, stored, and analysed through different stages to generate meaningful business insights and visual dashboards.

- **Milestones with Timelines**

Milestone ID	Milestone	Description	Start (Day)	End (Day)	Duration	Week(s)	Deliverables / Outcome
M1	 Data Collection & Cleaning	Collect raw data and basic cleaning	Day 1	Day 5	5 days	Week 1	Raw datasets collected and initial cleaned data
M2	 Data Cleaning & Preprocessing	Clean, validate and preprocess data	Day 6	Day 10	5 days	Week 2	Cleaned and preprocessed datasets
M3	 Data Transformation	Transform and prepare data for modeling	Day 11	Day 15	5 days	Week 3	Transformed data ready for ETL
M4	 ETL Development	Build ETL pipelines for data ingestion	Day 16	Day 20	5 days	Week 4	ETL pipelines developed and tested
M5	 Workflow Automation	Automate workflows using orchestration (Airflow)	Day 21	Day 25	5 days	Week 5	Automated workflows scheduled and running
M6	 Warehouse Design	Design and implement data warehouse	Day 26	Day 30	5 days	Week 6	Data warehouse built and populated
M7	 Business Insights Generation	Generate business insights and KPIs (SQL)	Day 31	Day 34	4 days	Week 6	Business KPIs and insights generated
M8	 Dashboard, Visualization, Testing & Deployment	Create dashboards, visualize insights, test and deploy the solution	Day 35	Day 42	8 days	Week 6	Dashboards created, system tested and deployed
M9	 Final Setup & Documentation	Final setup and project documentation	Day 43	Day 45	3 days	Week 6	Final documentation and handover

The milestones with timelines section gives a clear breakdown of the important phases of the End-to-End E-Commerce Data Platform for Sales and Customer Analytics project and when each phase is expected to be completed. It helps in tracking progress step by step, from data collection and processing to analytics, dashboard creation, testing, and final deployment, making the overall project execution more organized and easier to manage.

- **Directed Acyclic Graph (DAG)**

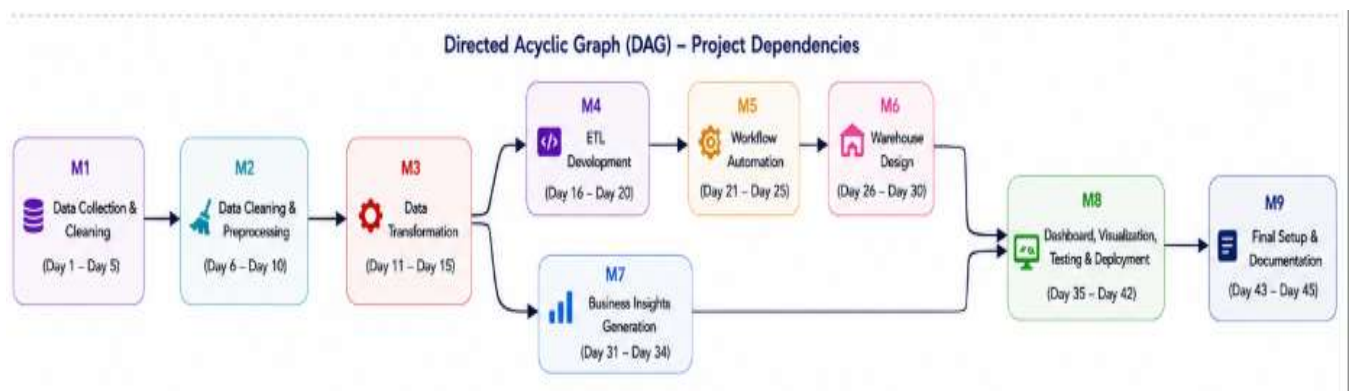


Fig 3: DAG

The Directed Acyclic Graph (DAG) shows how the tasks in the End-to-End E-Commerce Data Platform for Sales and Customer Analytics project are connected and executed in the correct order. For instance, data cleaning and preprocessing can only begin after the raw e-commerce data has been collected. Similarly, ETL pipeline development depends on the completion of data transformation, while dashboard creation depends on the successful setup of the data warehouse and analytics generation. At the same time, certain tasks such as workflow monitoring and documentation can be carried out independently alongside other activities. This flow helps the team manage the project in a smooth, organized, and efficient manner.

6. Tools and Technologies

- **Python**

Used for data cleaning, preprocessing, ETL scripting, and automation tasks.

- **Pandas & NumPy**

Used for handling datasets, performing data transformation, and analysis.

- **Apache NiFi**

Used for data ingestion, data routing, and managing the movement of raw data between different systems.

- **Apache Airflow**

Used for workflow scheduling, DAG management, and automating ETL pipelines.

- **Apache Spark**

Used for scalable and large-scale data processing and analytics.

- **PostgreSQL**

Used as the data warehouse for storing cleaned and transformed data.

- **SQL**

Used for querying the database and generating business insights.

- **Docker**

Used for containerization and maintaining a consistent development environment.

- **Power BI / Tableau**

Used for creating dashboards and visualizing sales and customer analytics.

- **Git & GitHub**

Used for version control, collaboration, and project repository management.

7. Cost Management

The End-to-End E-Commerce Data Platform for Sales and Customer Analytics project is designed in a cost-effective manner by mainly using free and open-source tools such as Python, Apache NiFi, Apache Airflow, PostgreSQL, Docker, and GitHub. Since these technologies are freely available, the project can be developed and tested on a normal personal computer without significant software or infrastructure costs. The overall setup mainly requires a system with moderate RAM, storage, and internet connectivity. By using workflow automation and open-source technologies, the project achieves efficient data processing and management while keeping the overall cost practical and affordable.